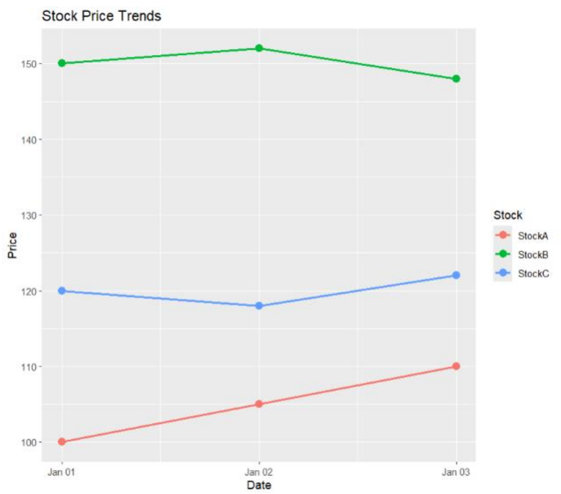
SET 20

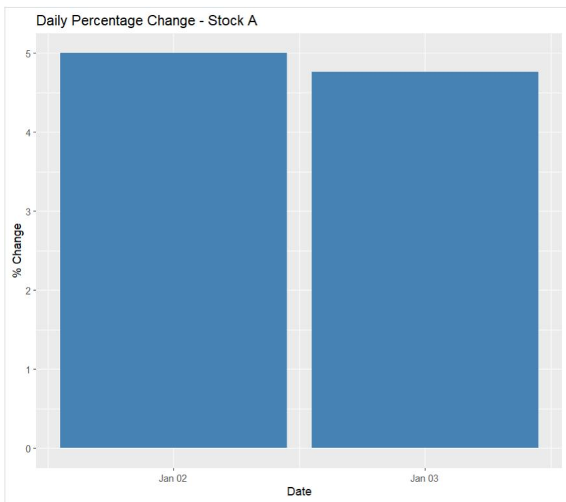
Stock Analysis

Dataset (CSV)

Procedure

1. Create the stock price dataset containing daily prices of three stocks.
2. Import the dataset into R Studio.
3. Load the necessary packages such as ggplot2, tidyr, dplyr, and DT.
4. Convert the data into long format for visualization.
5. Generate a line chart showing stock price trends over time.
6. Calculate daily percentage changes for Stock A.
7. Create a bar chart representing daily percentage changes.
8. Build an interactive table displaying stock prices.
9. Integrate the charts and table into a Tableau dashboard.
10. Interpret stock movement trends and identify periods of growth or decline.





R Code

```
library(ggplot2)
library(tidyr)
```

```
library(dplyr)
library(DT)

stocks <- data.frame(
 Date=as.Date(c("2023-01-01",
                "2023-01-02",
                "2023-01-03")),
 StockA=c(100,105,110),
 StockB=c(150,152,148),
 StockC=c(120,118,122)
)

# Q1 Line Chart
stock_long <- pivot_longer(
 stocks,
 cols=c(StockA,StockB,StockC),
 names_to="Stock",
 values_to="Price"
)

ggplot(stock_long,
       aes(Date,Price,color=Stock))+
  geom_line(size=1)+
  geom_point(size=3)+
  labs(title="Stock Price Trends",
       x="Date",
       y="Price")

# Q2 Percentage Change for Stock A
stocks$PctChange <- c(
  NA,
  diff(stocks$StockA) /
    head(stocks$StockA,-1) * 100
)

ggplot(na.omit(stocks),
       aes(Date,PctChange))+
```

```
  geom_bar(stat="identity",
           fill="steelblue")+
  labs(title="Daily Percentage Change - Stock A",
       x="Date",
       y="% Change")

# Q3 Table
datatable(stocks,
          caption="Stock Price Data")

# Q4 Dashboard Components
# Line Chart + Percentage Change Bar Chart + Table
```